Practical No. 1 – Searching and Sorting using C++

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Problem Statement : Searching and Sorting -- CO1, CO2

Design a program to maintain a student database that performs the following tasks:

1. Add and store student details (ID, Name, CGPA) using dynamically allocated memory.
2. Expand the student list using realloc() as new entries are added.
3. Implement Linear Search and Binary Search to find student records by ID.
4. Implement at least two Sorting Algorithms (Bubble Sort, Selection Sort, or Insertion Sort) to sort student records by: a. Name (Alphabetically) b. CGPA (Ascending/Descending)
5. Analyze and compare the performance of search operations before and after sorting.

Code

```
#include <iostream>
#include <algorithm> using
namespace std;
struct Student
{ int
id;
string name; float
cgpa;
};
int linearSearch(Student s[], int n, int key)
{
for(int i=0;i<n;i++)
{
```

```
if(s[i].id==key)
return i;
}
return -1;
}
int binarySearch(Student s[], int n, int key)
{
int low=0; int high=n-1;
while(low<=high) { int
mid=(low+high)/2; if(s[mid].id==key)
return mid; if(key<s[mid].id) high=mid-
1; else low=mid+1; } return -1;} void
bubbleSortName(Student s[],int n)
{
for(int i=0;i<n-1;i++)
{
for(int j=0;j<n-i-1;j++)
{
if(s[j].name>s[j+1].name) swap(s[j],s[j+1]);
}
}
}
void selectionSortCGPA(Student s[],int n)
{
for(int i=0;i<n-1;i++)
```

```
{
int min=i; for(int
j=i+1;j<n;j++) {
if(s[j].cgpa<s[min].cgpa)
min=j;
}
swap(s[i],s[min]);
}
}
int main()
{ int n; cout<<"Enter Number of
Students : "; cin>>n; Student *s=new
Student[n]; for(int i=0;i<n;i++)
{
cout<<"\nStudent "<<i+1<<endl;
cout<<"ID : "; cin>>s[i].id;
cout<<"Name : ";
cin>>s[i].name; cout<<"CGPA : ";
cin>>s[i].cgpa;
}
cout<<"\nStudent Records\n"; for(int i=0;i<n;i++)
{
cout<<s[i].id<<" " <<s[i].name<<"
"
<<s[i].cgpa<<endl;
```

```
}
int key; cout<<"\nEnter ID to Search : ";
cin>>key; int pos=linearSearch(s,n,key);
if(pos!=-1) cout<<"Found using Linear
Search"<<endl; else cout<<"Not
Found"<<endl; bubbleSortName(s,n);
cout<<"\nSorted by Name\n";
for(int i=0;i<n;i++)
{
cout<<s[i].id<<" " <<s[i].name<<"
"
<<s[i].cgpa<<endl;
}
selectionSortCGPA(s,n); cout<<"\nSorted
by CGPA\n";
for(int i=0;i<n;i++) {
cout<<s[i].id<<"    "
<<s[i].name<<" "
<<s[i].cgpa<<endl;
}
sort(s,s+n,[](Student a,Student b)
{
return a.id<b.id;
});
```

```
pos=binarySearch(s,n,key); if(pos!=-1)
cout<<"\nFound using Binary Search"<<endl;
else cout<<"\nNot Found"<<endl;
delete[] s; return
0;
}
```

Output

```
 Enter Number of Students : 3
Student 1
ID : 215
Name : Sneha
CGPA : 9.18
Student 2
ID : 216
Name : Anu
CGPA : 8.4

Student 3
ID : 217
Name : Shrutika
CGPA : 8.91
```

Student Records

```
215 Sneha 9.18
216 Anu 8.4
```

217 Shrutika 8.91

Enter ID to Search : 240

Found using Linear Search

Sorted by Name

217 Shrutika 8.91

215 Sneha 9.18

216 Anu 8.4

Sorted by CGPA

216 Anu 8.4

217 Shrutika 8.91

215 Sneha 9.18